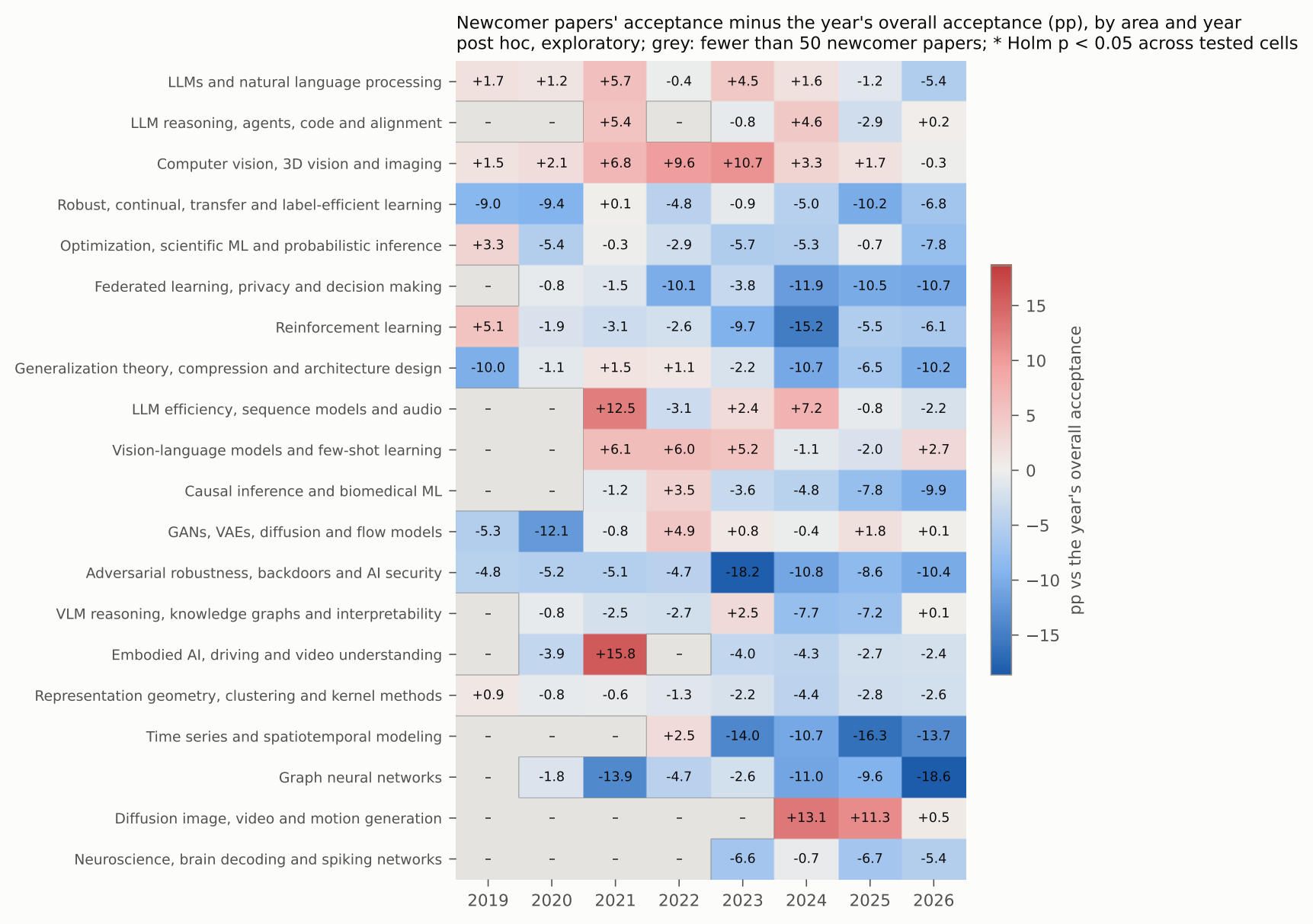
Newcomer papers' acceptance minus the year's overall acceptance (pp), by area and year
post hoc, exploratory; grey: fewer than 50 newcomer papers; * Holm p < 0.05 across tested cells
LLMs and natural language processing +1.7 +1.2 +5.7 -0.4 +4.5 +1.6 -1.2 -5.4
LLM reasoning, agents, code and alignment – – +5.4 – -0.8 +4.6 -2.9 +0.2
Computer vision, 3D vision and imaging +1.5 +2.1 +6.8 +9.6 +10.7 +3.3 +1.7 -0.3
Robust, continual, transfer and label-efficient learning -9.0 -9.4 +0.1 -4.8 -0.9 -5.0 -10.2 -6.8
Optimization, scientific ML and probabilistic inference +3.3 -5.4 -0.3 -2.9 -5.7 -5.3 -0.7 -7.8
Federated learning, privacy and decision making – -0.8 -1.5 -10.1 -3.8 -11.9 -10.5 -10.7
Reinforcement learning +5.1 -1.9 -3.1 -2.6 -9.7 -15.2 -5.5 -6.1
Generalization theory, compression and architecture design -10.0 -1.1 +1.5 +1.1 -2.2 -10.7 -6.5 -10.2
LLM efficiency, sequence models and audio – – +12.5 -3.1 +2.4 +7.2 -0.8 -2.2
Vision-language models and few-shot learning – – +6.1 +6.0 +5.2 -1.1 -2.0 +2.7
Causal inference and biomedical ML – – -1.2 +3.5 -3.6 -4.8 -7.8 -9.9
GANs, VAEs, diffusion and flow models -5.3 -12.1 -0.8 +4.9 +0.8 -0.4 +1.8 +0.1
Adversarial robustness, backdoors and AI security -4.8 -5.2 -5.1 -4.7 -18.2 -10.8 -8.6 -10.4
VLM reasoning, knowledge graphs and interpretability – -0.8 -2.5 -2.7 +2.5 -7.7 -7.2 +0.1
Embodied AI, driving and video understanding – -3.9 +15.8 – -4.0 -4.3 -2.7 -2.4
Representation geometry, clustering and kernel methods +0.9 -0.8 -0.6 -1.3 -2.2 -4.4 -2.8 -2.6
Time series and spatiotemporal modeling – – – +2.5 -14.0 -10.7 -16.3 -13.7
Graph neural networks – -1.8 -13.9 -4.7 -2.6 -11.0 -9.6 -18.6
Diffusion image, video and motion generation – – – – – +13.1 +11.3 +0.5
Neuroscience, brain decoding and spiking networks – – – – -6.6 -0.7 -6.7 -5.4
2019 2020 2021 2022 2023 2024 2025 2026
15 10 5 0 −5 −10 −15
pp vs the year's overall acceptance